In [1]:
```
# ============================================================
# MEMBER 4
# VISUALIZATION & INTERPRETATION
# STEPS 57-70
# ============================================================

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
import sys

from IPython.display import display

%matplotlib inline

pd.set_option(
    "display.max_columns",
    None
)

pd.set_option(
    "display.width",
    200
)


# ============================================================
# FILE PATHS
# ============================================================

INPUT_FILE = (
    r"C:\Users\admin\OneDrive\Desktop\DS- Activity 1\backblaze_analytical_datase
)

OUTPUT_FOLDER = (
    "output/visualizations"
)

os.makedirs(
    OUTPUT_FOLDER,
    exist_ok=True
)


# ============================================================
# HELPER FUNCTION
# JUPYTER + TERMINAL + CSV
# ============================================================

def show_and_save(
    data,
    filename,
    title
):

    print("\n" + "=" * 70)
```

```
    print(title)
    print("=" * 70)

    # JUPYTER
    display(data)

    # TERMINAL
    try:

        sys.__stdout__.write(
            "\n" +
            "=" * 70 +
            "\n" +
            title +
            "\n" +
            "=" * 70 +
            "\n" +
            data.to_string(
                index=False
            ) +
            "\n"
        )

        sys.__stdout__.flush()

    except Exception:
        pass

    # CSV
    filepath = os.path.join(
        OUTPUT_FOLDER,
        filename
    )

    data.to_csv(
        filepath,
        index=False
    )

    print(
        "\nSaved CSV:"
    )

    print(
        filepath
    )

    return data
```

In [2]:

```
# ============================================================
# STEP 57 — LOAD CLEANED DATASET
# ============================================================

print("\n" + "=" * 70)
print("STEP 57 — LOAD CLEANED DATASET")
print("=" * 70)

df = pd.read_csv(
    INPUT_FILE
)
```

```
print(
    "Rows:",
    df.shape[0]
)

print(
    "Columns:",
    df.shape[1]
)

try:

    sys.__stdout__.write(
        f"\nSTEP 57 — Rows: {df.shape[0]}, "
        f"Columns: {df.shape[1]}\n"
    )

    sys.__stdout__.flush()

except Exception:
    pass
```

```
======================================================================
STEP 57 — LOAD CLEANED DATASET
======================================================================
Rows: 30943146
Columns: 19
```

In [3]:

```
# ============================================================
# STEP 58 — IDENTIFY COLUMNS
# ============================================================

print("\n" + "=" * 70)
print("STEP 58 — COLUMN IDENTIFICATION")
print("=" * 70)

numeric_columns = df.select_dtypes(
    include=np.number
).columns.tolist()

categorical_columns = df.select_dtypes(
    exclude=np.number
).columns.tolist()

print(
    "\nNumerical Columns:"
)

print(
    numeric_columns
)

print(
    "\nCategorical Columns:"
)

print(
    categorical_columns
)
```

```
# ============================================================
# TARGET COLUMN
# ============================================================

possible_targets = [

    "failure",

    "failed",

    "disk_failure",

    "disk_failed",

    "failure_status",

    "target",

    "label"
]

target_column = None

for column in possible_targets:

    if column in df.columns:

        target_column = column

        break


print(
    "\nTarget Column:",
    target_column
)
```

```
======================================================================
STEP 58 — COLUMN IDENTIFICATION
======================================================================

Numerical Columns:
['capacity_bytes', 'failure', 'smart_5_normalized', 'smart_5_raw', 'smart_9_norma
lized', 'smart_9_raw', 'smart_90_normalized', 'smart_90_raw', 'smart_187_normaliz
ed', 'smart_187_raw', 'smart_188_normalized', 'smart_188_raw', 'smart_197_normali
zed', 'smart_197_raw', 'smart_198_normalized', 'smart_198_raw']

Categorical Columns:
['date', 'serial_number', 'model']

Target Column: failure
```

In [4]:

```
# ============================================================
# STEP 59 — FEATURE SELECTION
# ============================================================

print("\n" + "=" * 70)
print("STEP 59 — FEATURE SELECTION")
```

```
print("=" * 70)

feature_columns = (
    numeric_columns.copy()
)

if target_column in feature_columns:

    feature_columns.remove(
        target_column
    )

selected_features = (
    feature_columns[:6]
)

selected_features_df = pd.DataFrame({

    "Selected_Feature":
        selected_features
})

show_and_save(
    selected_features_df,
    "step_59_selected_features.csv",
    "STEP 59 — SELECTED FEATURES"
)
```

```
======================================================================
STEP 59 — FEATURE SELECTION
======================================================================

======================================================================
STEP 59 — SELECTED FEATURES
======================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Selected_Feature</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> </tbody> <tbody></tbody>

	Selected_Feature
0	capacity_bytes
1	smart_5_normalized
2	smart_5_raw
3	smart_9_normalized
4	smart_9_raw
5	smart_90_normalized

```
Saved CSV:
output/visualizations\step_59_selected_features.csv
```

Out[4]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Selected_Feature</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> </tbody> <tbody></tbody>

capacity_bytes
smart_5_normalized
smart_5_raw
smart_9_normalized
smart_9_raw
smart_90_normalized

In [5]:
```
# ============================================================
# STEP 60 — BAR CHART
# ============================================================

print("\n" + "=" * 70)
print("STEP 60 — BAR CHART")
print("=" * 70)

if target_column:

    target_counts = (
        df[target_column]
        .value_counts()
        .sort_index()
    )

    target_result = pd.DataFrame({

        "Category":
            target_counts
            .index
            .astype(str),

        "Count":
            target_counts
            .values
    })

    show_and_save(
        target_result,
        "step_60_bar_chart_data.csv",
        "STEP 60 — BAR CHART DATA"
    )

    plt.figure(
        figsize=(8, 6)
    )

    plt.bar(
        target_result["Category"],
        target_result["Count"]
    )
```

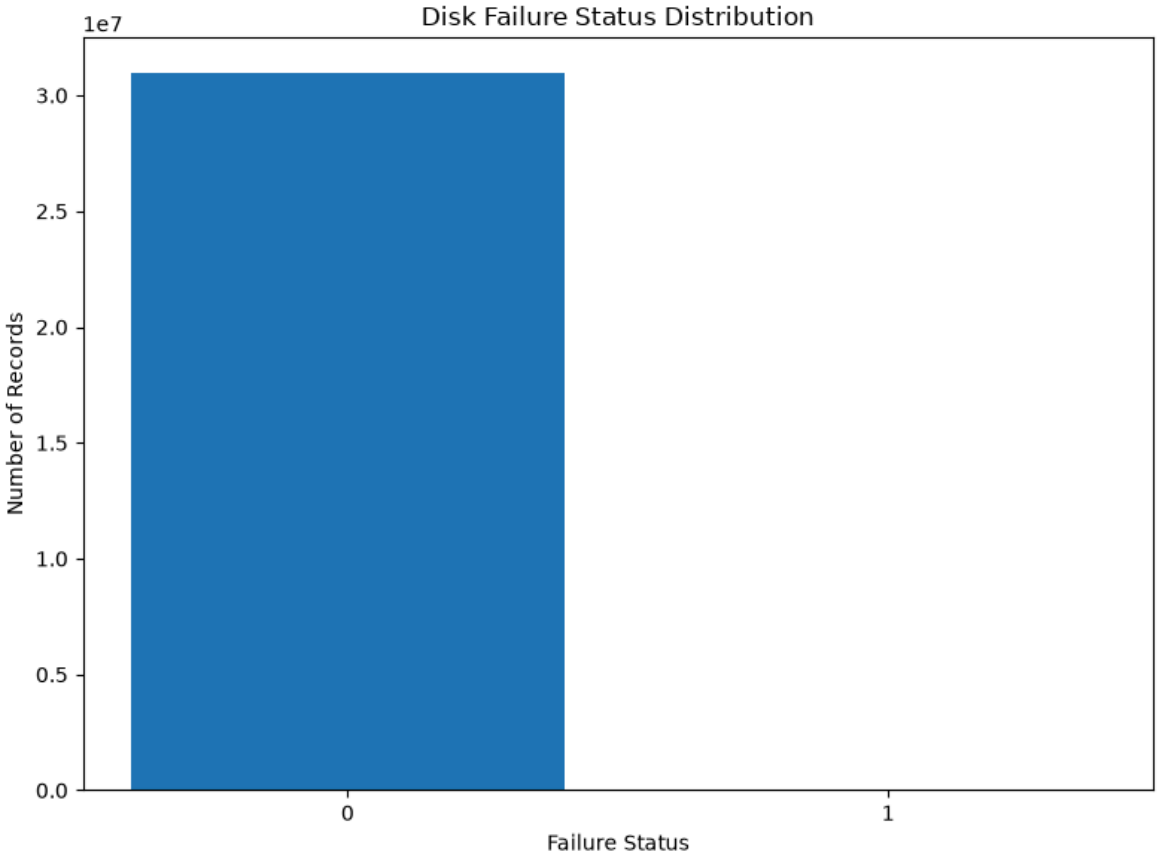
```
plt.title(
    "Disk Failure Status Distribution"
)

plt.xlabel(
    "Failure Status"
)

plt.ylabel(
    "Number of Records"
)

plt.xticks(
    rotation=0
)

plt.tight_layout()

graph_path = os.path.join(
    OUTPUT_FOLDER,
    "step_60_bar_chart.png"
)

plt.savefig(
    graph_path,
    dpi=300,
    bbox_inches="tight"
)

# JUPYTER
plt.show()

plt.close()

largest_category = (
    target_counts.idxmax()
)

largest_count = (
    target_counts.max()
)

interpretation = (

    f"The bar chart shows the distribution "
    f"of disk failure status. The category "
    f"'{largest_category}' has the highest "
    f"number of observations with "
    f"{largest_count} records."
)

print(
    "\nInterpretation:"
)

print(
    interpretation
)
```

```
interpretation_df = pd.DataFrame({

    "Step": [60],

    "Graph": [
        "Bar Chart"
    ],

    "Interpretation": [
        interpretation
    ]
})

interpretation_df.to_csv(
    os.path.join(
        OUTPUT_FOLDER,
        "step_60_bar_chart_interpretation.csv"
    ),
    index=False
)

print(
    "\nGraph saved:",
    graph_path
)
```

```
======================================================================
STEP 60 — BAR CHART
======================================================================

======================================================================
STEP 60 — BAR CHART DATA
======================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Category</th> <th>Count</th> </thead> <tbody> <th>0</th> <th>1</th> </tbody> <tbody></tbody>

```
0   30942092
1       1054

Saved CSV:
output/visualizations\step_60_bar_chart_data.csv
```



Interpretation:
The bar chart shows the distribution of disk failure status. The category '0' has the highest number of observations with 30942092 records.

Graph saved: output/visualizations\step_60_bar_chart.png

In [6]:
```
# ============================================================
# STEP 61 — LINE CHART
# ============================================================

print("\n" + "=" * 70)
print("STEP 61 — LINE CHART")
print("=" * 70)

feature = selected_features[0]

datetime_columns = []

for column in df.columns:

    column_lower = column.lower()

    if any(
        keyword in column_lower
        for keyword in [
            "date",
            "time",
            "timestamp"
        ]
    ):

        datetime_columns.append(
            column
        )
```

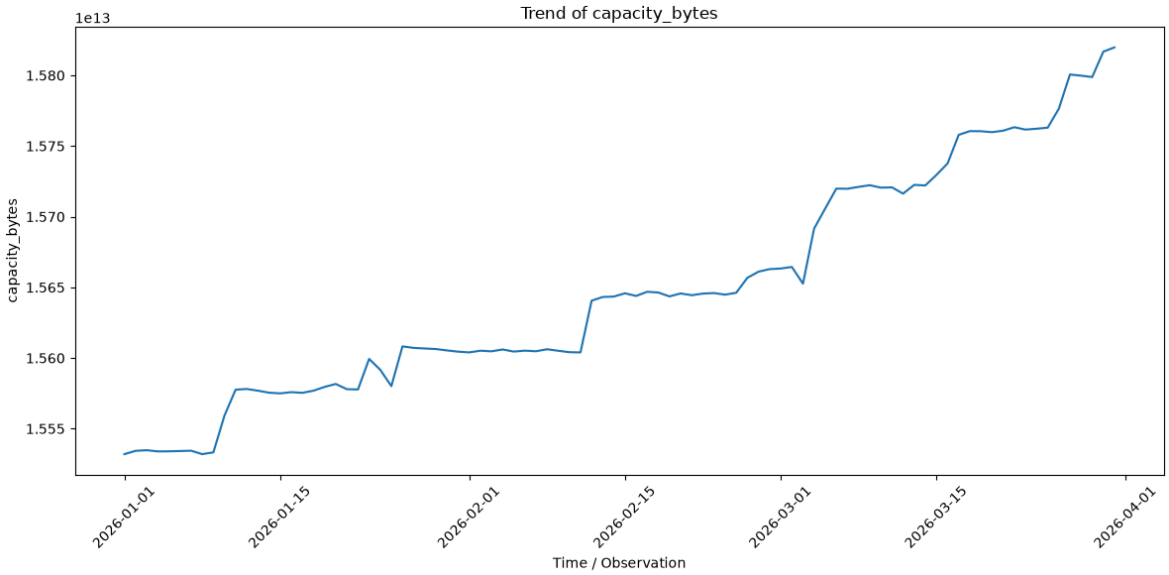
```
if len(datetime_columns) > 0:

    date_column = (
        datetime_columns[0]
    )

    temp_df = df.copy()

    temp_df[date_column] = (
        pd.to_datetime(
            temp_df[date_column],
            errors="coerce"
        )
    )

    temp_df = temp_df.dropna(
        subset=[
            date_column
        ]
    )

    if len(temp_df) > 0:

        trend_data = (

            temp_df

            .set_index(
                date_column
            )[feature]

            .resample(
                "D"
            )

            .mean()

            .dropna()
        )

        x_values = (
            trend_data.index
        )

        y_values = (
            trend_data.values
        )

    else:

        trend_data = (
            df[feature]
            .head(100)
            .reset_index(
                drop=True
            )
        )
```

```
        x_values = (
            trend_data.index
        )

        y_values = (
            trend_data.values
        )

else:

    trend_data = (
        df[feature]
        .head(100)
        .reset_index(
            drop=True
        )
    )

    x_values = (
        trend_data.index
    )

    y_values = (
        trend_data.values
    )


plt.figure(
    figsize=(12, 6)
)

plt.plot(
    x_values,
    y_values
)

plt.title(
    f"Trend of {feature}"
)

plt.xlabel(
    "Time / Observation"
)

plt.ylabel(
    feature
)

plt.xticks(
    rotation=45
)

plt.tight_layout()

graph_path = os.path.join(
    OUTPUT_FOLDER,
    "step_61_line_chart.png"
)

plt.savefig(
```

```
    graph_path,
    dpi=300,
    bbox_inches="tight"
)

plt.show()

plt.close()

if len(y_values) >= 2:

    if y_values[-1] > y_values[0]:

        trend_direction = (
            "increasing"
        )

    elif y_values[-1] < y_values[0]:

        trend_direction = (
            "decreasing"
        )

    else:

        trend_direction = (
            "relatively stable"
        )

else:

    trend_direction = (
        "not enough observations"
    )


interpretation = (

    f"The line chart shows the variation "
    f"of {feature}. The overall pattern "
    f"is {trend_direction}."
)

print(
    "\nInterpretation:"
)

print(
    interpretation
)

line_result = pd.DataFrame({

    "Feature": [
        feature
    ],

    "Trend": [
        trend_direction
    ],
```

```
    "Interpretation": [
        interpretation
    ]
})

show_and_save(
    line_result,
    "step_61_line_chart_result.csv",
    "STEP 61 — LINE CHART RESULT"
)

print(
    "\nGraph saved:",
    graph_path
)
```

```
========================================================================
STEP 61 — LINE CHART
========================================================================
```




```
Interpretation:
The line chart shows the variation of capacity_bytes. The overall pattern is incr
easing.

========================================================================
STEP 61 — LINE CHART RESULT
========================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Feature</th> <th>Trend</th> <th>Interpretation</th> </thead> <tbody> <th>0</th> </tbody> <tbody></tbody>

	Feature	Trend	Interpretation
0	capacity_bytes	increasing	The line chart shows the variation of capacity...

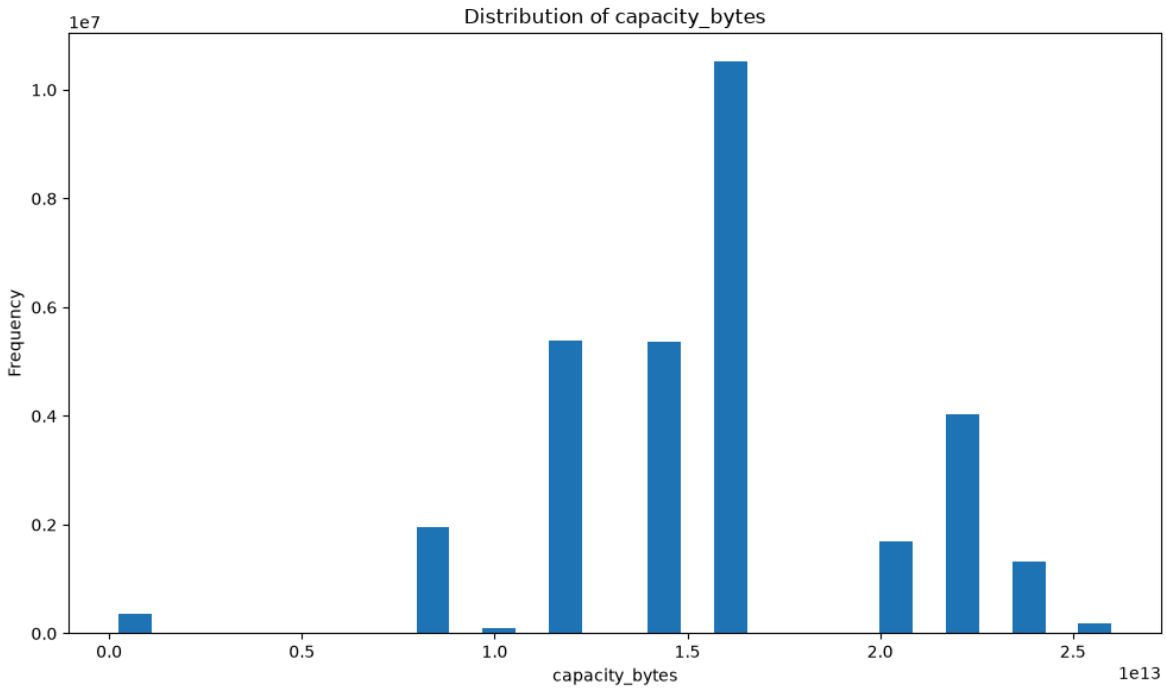
```
Saved CSV:
output/visualizations\step_61_line_chart_result.csv

Graph saved: output/visualizations\step_61_line_chart.png
```

In [7]:
```
# ============================================================
# STEP 62 — HISTOGRAM
# ============================================================
```

```
print("\n" + "=" * 70)
print("STEP 62 — HISTOGRAM")
print("=" * 70)

feature = (
    selected_features[0]
)

skew_value = (
    df[feature].skew()
)

plt.figure(
    figsize=(10, 6)
)

plt.hist(
    df[feature].dropna(),
    bins=30
)

plt.title(
    f"Distribution of {feature}"
)

plt.xlabel(
    feature
)

plt.ylabel(
    "Frequency"
)

plt.tight_layout()

graph_path = os.path.join(
    OUTPUT_FOLDER,
    "step_62_histogram.png"
)

plt.savefig(
    graph_path,
    dpi=300,
    bbox_inches="tight"
)

plt.show()

plt.close()


if skew_value > 1:

    distribution_type = (
        "Highly positively skewed"
    )

elif skew_value > 0.5:
```

```
    distribution_type = (
        "Moderately positively skewed"
    )

elif skew_value < -1:

    distribution_type = (
        "Highly negatively skewed"
    )

elif skew_value < -0.5:

    distribution_type = (
        "Moderately negatively skewed"
    )

else:

    distribution_type = (
        "Approximately symmetric"
    )


interpretation = (

    f"The histogram shows the distribution "
    f"of {feature}. The distribution is "
    f"{distribution_type}, with a skewness "
    f"value of {skew_value:.2f}."
)

hist_result = pd.DataFrame({

    "Feature": [
        feature
    ],

    "Skewness": [
        skew_value
    ],

    "Distribution": [
        distribution_type
    ],

    "Interpretation": [
        interpretation
    ]
})

show_and_save(
    hist_result,
    "step_62_histogram_result.csv",
    "STEP 62 — HISTOGRAM RESULT"
)

print(
    "\nGraph saved:",
    graph_path
)
```

```
======================================================================
STEP 62 — HISTOGRAM
======================================================================
```




```
======================================================================
STEP 62 — HISTOGRAM RESULT
======================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Feature</th> <th>Skewness</th> <th>Distribution</th> <th>Interpretation</th> </thead> <tbody> <th>0</th> </tbody> <tbody></tbody>

capacity_bytes	-0.130304	Approximately symmetric	The histogram shows the distribution of capaci...	

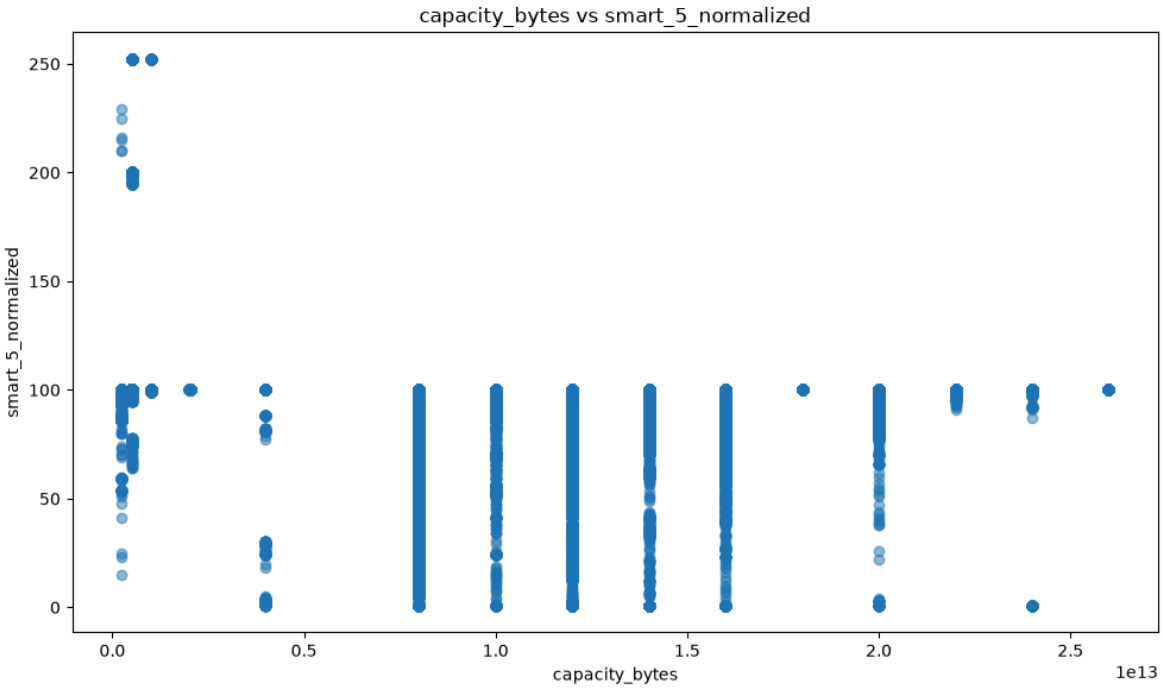
```
Saved CSV:
output/visualizations\step_62_histogram_result.csv

Graph saved: output/visualizations\step_62_histogram.png
```

In [8]:
```python
# ============================================================
# STEP 63 — SCATTER PLOT
# ============================================================

print("\n" + "=" * 70)
print("STEP 63 — SCATTER PLOT")
print("=" * 70)

if len(selected_features) >= 2:

    feature_x = (
        selected_features[0]
    )

    feature_y = (
        selected_features[1]
    )

    correlation = (
```

```
    df[
        [
            feature_x,
            feature_y
        ]
    ]
    .corr()
    .iloc[0, 1]
)

plt.figure(
    figsize=(10, 6)
)

plt.scatter(
    df[feature_x],
    df[feature_y],
    alpha=0.5
)

plt.title(
    f"{feature_x} vs {feature_y}"
)

plt.xlabel(
    feature_x
)

plt.ylabel(
    feature_y
)

plt.tight_layout()

graph_path = os.path.join(
    OUTPUT_FOLDER,
    "step_63_scatter_plot.png"
)

plt.savefig(
    graph_path,
    dpi=300,
    bbox_inches="tight"
)

plt.show()

plt.close()


if abs(correlation) >= 0.7:

    relationship = "Strong"

elif abs(correlation) >= 0.4:

    relationship = "Moderate"

elif abs(correlation) >= 0.2:
```

```
        relationship = "Weak"

    else:

        relationship = (
            "Very weak / negligible"
        )


    interpretation = (

        f"The scatter plot compares "
        f"{feature_x} and {feature_y}. "
        f"The correlation is "
        f"{correlation:.2f}, indicating a "
        f"{relationship.lower()} relationship."
    )


    scatter_result = pd.DataFrame({

        "Feature_X": [
            feature_x
        ],

        "Feature_Y": [
            feature_y
        ],

        "Correlation": [
            correlation
        ],

        "Relationship": [
            relationship
        ],

        "Interpretation": [
            interpretation
        ]
    })


    show_and_save(
        scatter_result,
        "step_63_scatter_result.csv",
        "STEP 63 — SCATTER PLOT RESULT"
    )

    print(
        "\nGraph saved:",
        graph_path
    )
```

```
======================================================================
STEP 63 — SCATTER PLOT
======================================================================
```




```
======================================================================
STEP 63 — SCATTER PLOT RESULT
======================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Feature_X</th> <th>Feature_Y</th> <th>Correlation</th> <th>Relationship</th> <th>Interpretation</th> </thead> <tbody> <th>0</th> </tbody> <tbody></tbody>

capacity_bytes	smart_5_normalized	-0.04467	Very weak / negligible	The scatter plot compares capacity_bytes and s...

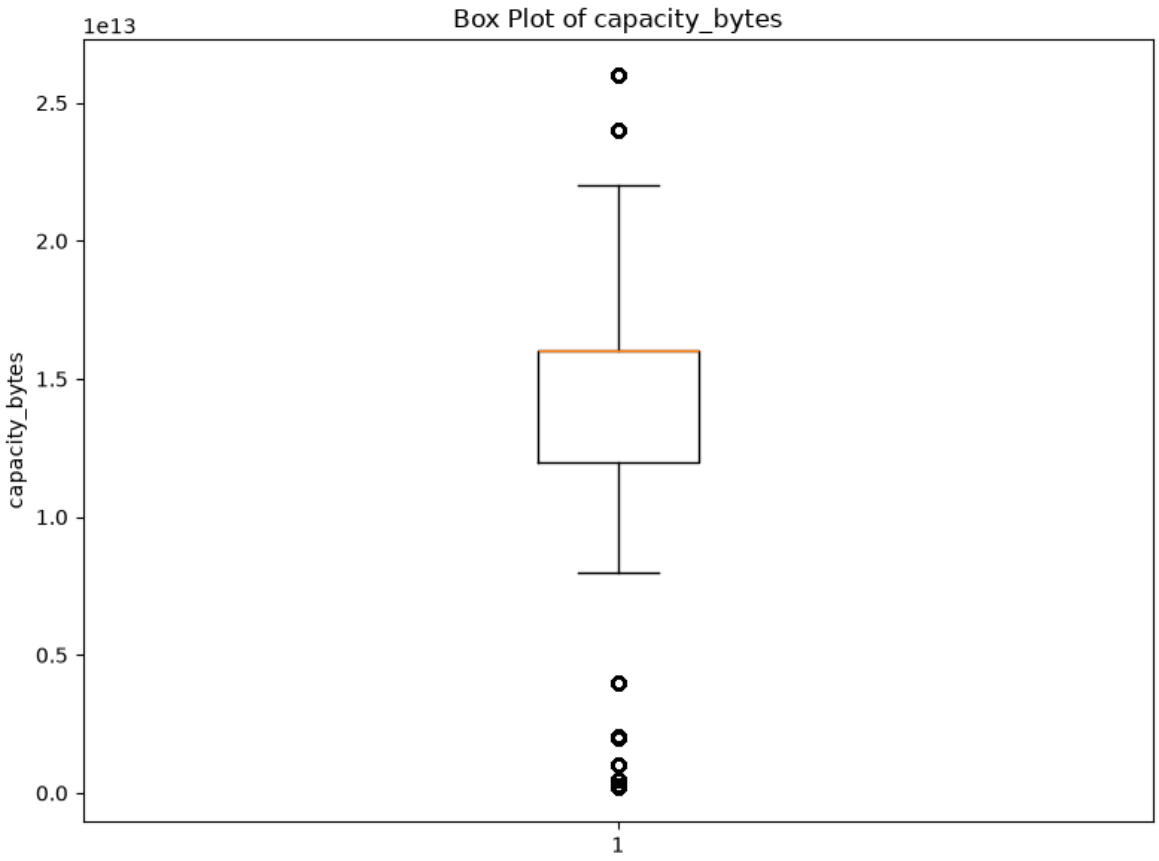
```
Saved CSV:
output/visualizations\step_63_scatter_result.csv

Graph saved: output/visualizations\step_63_scatter_plot.png
```

In [9]:
```python
# ============================================================
# STEP 64 — BOX PLOT
# ============================================================

print("\n" + "=" * 70)
print("STEP 64 — BOX PLOT")
print("=" * 70)

feature = (
    selected_features[0]
)

Q1 = (
    df[feature]
    .quantile(0.25)
)

Q3 = (
    df[feature]
    .quantile(0.75)
)
```

```
IQR = (
    Q3 - Q1
)

lower_bound = (
    Q1 - 1.5 * IQR
)

upper_bound = (
    Q3 + 1.5 * IQR
)

outliers = df[
    (df[feature] < lower_bound)
    |
    (df[feature] > upper_bound)
]


plt.figure(
    figsize=(8, 6)
)

plt.boxplot(
    df[feature].dropna()
)

plt.title(
    f"Box Plot of {feature}"
)

plt.ylabel(
    feature
)

plt.tight_layout()

graph_path = os.path.join(
    OUTPUT_FOLDER,
    "step_64_box_plot.png"
)

plt.savefig(
    graph_path,
    dpi=300,
    bbox_inches="tight"
)

plt.show()

plt.close()


box_result = pd.DataFrame({

    "Feature": [
        feature
    ],
```

```
    "Q1": [
        Q1
    ],

    "Q3": [
        Q3
    ],

    "IQR": [
        IQR
    ],

    "Lower_Bound": [
        lower_bound
    ],

    "Upper_Bound": [
        upper_bound
    ],

    "Potential_Outliers": [
        len(outliers)
    ]
})


show_and_save(
    box_result,
    "step_64_box_plot_result.csv",
    "STEP 64 — BOX PLOT RESULT"
)

print(
    "\nInterpretation:"
)

print(

    f"The box plot shows the spread, median "
    f"and quartiles of {feature}. "
    f"There are approximately "
    f"{len(outliers):,} potential outliers "
    f"using the IQR method."
)

print(
    "\nGraph saved:",
    graph_path
)
```

```
======================================================================
STEP 64 — BOX PLOT
======================================================================
```




```
======================================================================
STEP 64 — BOX PLOT RESULT
======================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Feature</th> <th>Q1</th> <th>Q3</th> <th>IQR</th> <th>Lower_Bound</th> <th>Upper_Bound</th> <th>Potential_Outliers</th> </thead> <tbody> <th>0</th> </tbody> <tbody></tbody>

	Feature	Q1	Q3	IQR	Lower_Bound	Upper_Bound	Potential_Outliers
0	capacity_bytes	1.200014e+13	1.600090e+13	4.000762e+12	5.998996e+12	2.200204e+13	18

```
Saved CSV:
output/visualizations\step_64_box_plot_result.csv

Interpretation:
The box plot shows the spread, median and quartiles of capacity_bytes. There are
approximately 1,892,988 potential outliers using the IQR method.

Graph saved: output/visualizations\step_64_box_plot.png
```

In [10]:
```
# ============================================================
# STEP 65 — PIE CHART
# ============================================================

print("\n" + "=" * 70)
print("STEP 65 — PIE CHART")
print("=" * 70)

if target_column:

    pie_data = (
        df[target_column]
```

```
    .value_counts()
)

pie_result = pd.DataFrame({

    "Category":
        pie_data.index.astype(str),

    "Count":
        pie_data.values,

    "Percentage":
        (
            pie_data.values
            /
            pie_data.values.sum()
            *
            100
        )
})


show_and_save(
    pie_result,
    "step_65_pie_chart_data.csv",
    "STEP 65 — PIE CHART DATA"
)


if len(pie_data) <= 6:

    plt.figure(
        figsize=(8, 8)
    )

    plt.pie(

        pie_data.values,

        labels=
            pie_data
            .index
            .astype(str),

        autopct="%1.1f%%",

        startangle=90
    )

    plt.title(
        "Disk Failure Status Proportion"
    )

    plt.tight_layout()

    graph_path = os.path.join(
        OUTPUT_FOLDER,
        "step_65_pie_chart.png"
    )
```

```
plt.savefig(
    graph_path,
    dpi=300,
    bbox_inches="tight"
)

plt.show()

plt.close()


largest_category = (
    pie_data.idxmax()
)

largest_percentage = (

    pie_data.max()
    /
    pie_data.sum()
    *
    100
)


interpretation = (

    f"The pie chart shows the proportion "
    f"of failure categories. "
    f"'{largest_category}' represents "
    f"the largest proportion at "
    f"{largest_percentage:.2f}%."
)


print(
    "\nInterpretation:"
)

print(
    interpretation
)


pie_interpretation = pd.DataFrame({

    "Step": [
        65
    ],

    "Graph": [
        "Pie Chart"
    ],

    "Interpretation": [
        interpretation
    ]
})
```

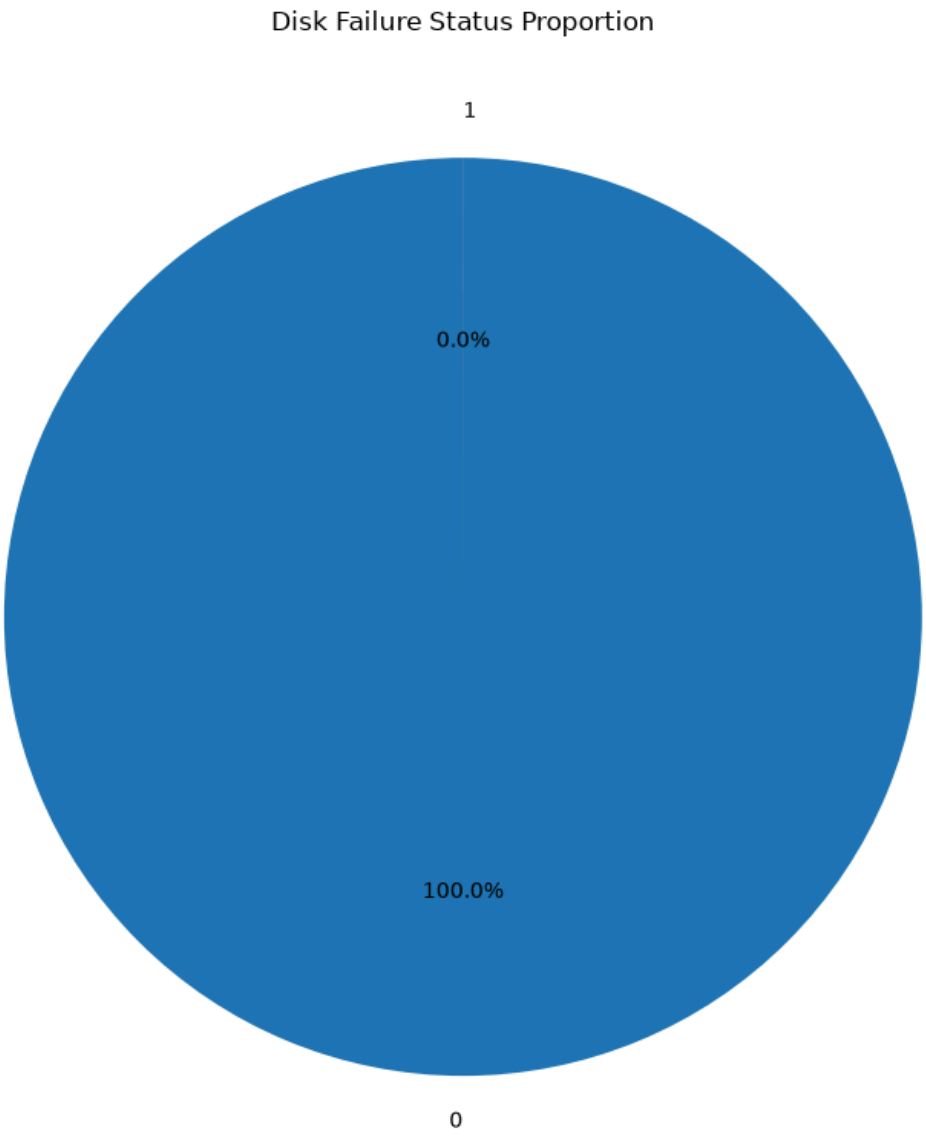
```
        pie_interpretation.to_csv(

            os.path.join(
                OUTPUT_FOLDER,
                "step_65_pie_chart_interpretation.csv"
            ),

            index=False
        )


        print(
            "\nGraph saved:",
            graph_path
        )

    else:

        print(
            "Pie chart skipped because there "
            "are more than 6 categories."
        )


else:

    print(
        "Target column was not detected."
    )
```

```
======================================================================
STEP 65 — PIE CHART
======================================================================

======================================================================
STEP 65 — PIE CHART DATA
======================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Category</th> <th>Count</th> <th>Percentage</th> </thead> <tbody> <th>0</th> <th>1</th> </tbody> <tbody></tbody>

0	30942092	99.996594	
1	1054	0.003406	

```
Saved CSV:
output/visualizations\step_65_pie_chart_data.csv
```




```
Interpretation:
The pie chart shows the proportion of failure categories. '0' represents the larg
est proportion at 100.00%.

Graph saved: output/visualizations\step_65_pie_chart.png
```

In [11]:
```python
# ============================================================
# STEP 66 — CORRELATION HEATMAP
# ============================================================

print("\n" + "=" * 70)
print("STEP 66 — CORRELATION HEATMAP")
print("=" * 70)

correlation_matrix = (
    df[numeric_columns]
    .corr()
)

display(
    correlation_matrix
)

print(
```

```
    correlation_matrix.to_string()
)

correlation_matrix.to_csv(

    os.path.join(
        OUTPUT_FOLDER,
        "step_66_correlation_matrix.csv"
    ),

    index=True
)


plt.figure(
    figsize=(12, 9)
)

sns.heatmap(

    correlation_matrix,

    annot=True,

    fmt=".2f",

    cmap="coolwarm",

    center=0
)

plt.title(
    "Correlation Heatmap of Numerical Features"
)

plt.tight_layout()

graph_path = os.path.join(
    OUTPUT_FOLDER,
    "step_66_correlation_heatmap.png"
)

plt.savefig(
    graph_path,
    dpi=300,
    bbox_inches="tight"
)

plt.show()

plt.close()


corr_pairs = []

for i in range(
    len(correlation_matrix.columns)
):

    for j in range(
```

```
        i + 1,
        len(correlation_matrix.columns)
    ):

        corr_value = (
            correlation_matrix.iloc[
                i,
                j
            ]
        )

        corr_pairs.append({

            "Feature_1":
                correlation_matrix.columns[i],

            "Feature_2":
                correlation_matrix.columns[j],

            "Correlation":
                corr_value,

            "Absolute_Correlation":
                abs(corr_value)
        })


corr_pairs_df = pd.DataFrame(
    corr_pairs
)


if len(corr_pairs_df) > 0:

    strongest = corr_pairs_df.loc[
        corr_pairs_df[
            "Absolute_Correlation"
        ].idxmax()
    ]

    interpretation = (

        f"The strongest relationship is between "
        f"{strongest['Feature_1']} and "
        f"{strongest['Feature_2']}, with a "
        f"correlation of "
        f"{strongest['Correlation']:.2f}."
    )

    print(
        "\nStrongest Correlation:"
    )

    display(
        strongest.to_frame(
            "Value"
        )
    )

    print(
```

```
        "\nInterpretation:"
    )

    print(
        interpretation
    )

    corr_pairs_df.to_csv(

        os.path.join(
            OUTPUT_FOLDER,
            "step_66_correlation_pairs.csv"
        ),

        index=False
    )

else:

    interpretation = (
        "The heatmap displays relationships "
        "among numerical features."
    )


print(
    "\nGraph saved:",
    graph_path
)
```

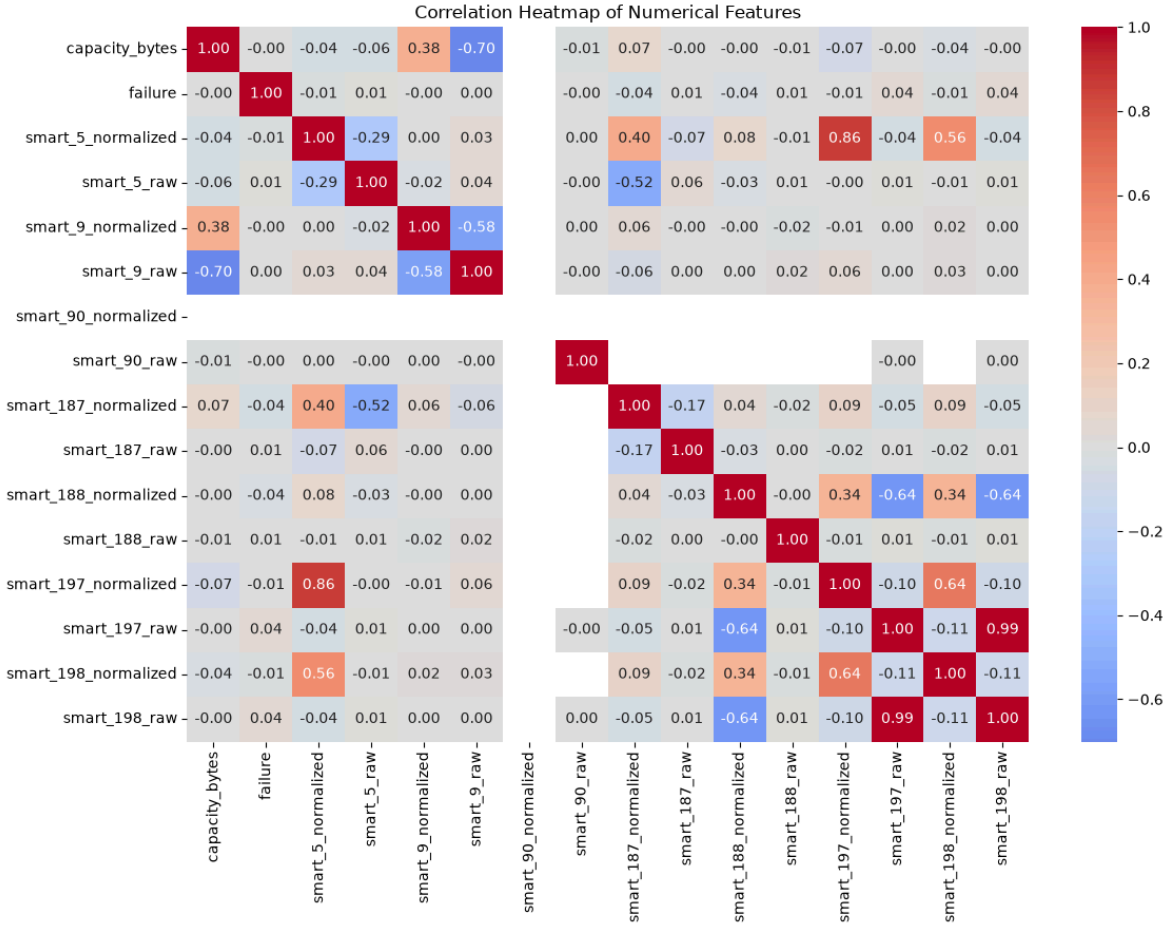
```
======================================================================
STEP 66 — CORRELATION HEATMAP
======================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>capacity_bytes</th> <th>failure</th> <th>smart_5_normalized</th> <th>smart_5_raw</th> <th>smart_9_normalized</th> <th>smart_9_raw</th> <th>smart_90_normalized</th> <th>smart_90_raw</th> <th>smart_187_normalized</th> <th>smart_187_raw</th> <th>smart_188_normalized</th> <th>smart_188_raw</th> <th>smart_197_normalized</th> <th>smart_197_raw</th> <th>smart_198_normalized</th> <th>smart_198_raw</th> </thead> <tbody> <th>capacity_bytes</th> <th>failure</th> <th>smart_5_normalized</th> <th>smart_5_raw</th> <th>smart_9_normalized</th> <th>smart_9_raw</th> <th>smart_90_normalized</th> <th>smart_90_raw</th> <th>smart_187_normalized</th> <th>smart_187_raw</th> <th>smart_188_normalized</th> <th>smart_188_raw</th> <th>smart_197_normalized</th> <th>smart_197_raw</th> <th>smart_198_normalized</th> <th>smart_198_raw</th> </tbody> <tbody></tbody>

1.000000	-0.001184	-0.044670	-0.055657	0.381113	-0.700703	NaN	-0.012504	0.071719
-0.001184	1.000000	-0.011817	0.014024	-0.000546	0.001993	NaN	-0.000035	-0.044238
-0.044670	-0.011817	1.000000	-0.291477	0.002428	0.034910	NaN	0.000001	0.397298
-0.055657	0.014024	-0.291477	1.000000	-0.024892	0.043867	NaN	-0.000085	-0.521230
0.381113	-0.000546	0.002428	-0.024892	1.000000	-0.581247	NaN	0.000715	0.062075
-0.700703	0.001993	0.034910	0.043867	-0.581247	1.000000	NaN	-0.003918	-0.061517
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
-0.012504	-0.000035	0.000001	-0.000085	0.000715	-0.003918	NaN	1.000000	NaN
0.071719	-0.044238	0.397298	-0.521230	0.062075	-0.061517	NaN	NaN	1.000000
-0.002513	0.009072	-0.070568	0.062572	-0.001613	0.001540	NaN	NaN	-0.165096
-0.001367	-0.036524	0.079715	-0.025987	-0.001278	0.001295	NaN	NaN	0.035482
-0.008700	0.006620	-0.007481	0.005007	-0.017820	0.017709	NaN	NaN	-0.015066
-0.072106	-0.007484	0.857067	-0.003841	-0.008305	0.055798	NaN	NaN	0.094061
-0.001853	0.039558	-0.044346	0.012066	0.002043	0.001423	NaN	-0.000050	-0.054375
-0.039339	-0.014159	0.560889	-0.005163	0.017945	0.028147	NaN	NaN	0.094061
-0.001760	0.037546	-0.043418	0.012174	0.000525	0.001436	NaN	0.000090	-0.054375

```
                     capacity_bytes   failure  smart_5_normalized  smart_5_raw
smart_9_normalized  smart_9_raw  smart_90_normalized  smart_90_raw  smart_187_nor
malized  smart_187_raw  smart_188_normalized  smart_188_raw  smart_197_normalized
smart_197_raw  smart_198_normalized  smart_198_raw
capacity_bytes             1.000000 -0.001184           -0.044670    -0.055657
0.381113    -0.700703                  NaN     -0.012504              0.071719
-0.002513             -0.001367      -0.008700             -0.072106      -0.0018
53             -0.039339      -0.001760
failure                   -0.001184  1.000000           -0.011817     0.014024
-0.000546     0.001993                  NaN     -0.000035             -0.044238
0.009072             -0.036524       0.006620             -0.007484       0.03955
8             -0.014159       0.037546
smart_5_normalized         -0.044670 -0.011817            1.000000    -0.291477
0.002428     0.034910                  NaN      0.000001              0.397298
-0.070568              0.079715      -0.007481              0.857067      -0.0443
46              0.560889      -0.043418
smart_5_raw                -0.055657  0.014024           -0.291477     1.000000
-0.024892     0.043867                  NaN     -0.000085             -0.521230
0.062572             -0.025987       0.005007             -0.003841       0.01206
6             -0.005163       0.012174
smart_9_normalized          0.381113 -0.000546            0.002428    -0.024892
1.000000    -0.581247                  NaN      0.000715              0.062075
-0.001613             -0.001278      -0.017820             -0.008305       0.0020
43              0.017945       0.000525
smart_9_raw                -0.700703  0.001993            0.034910     0.043867
-0.581247     1.000000                  NaN     -0.003918             -0.061517
0.001540              0.001295       0.017709              0.055798       0.00142
3              0.028147       0.001436
smart_90_normalized              NaN       NaN                NaN          NaN
NaN          NaN                  NaN           NaN                    NaN
NaN                   NaN            NaN                   NaN            NaN
NaN            NaN
smart_90_raw               -0.012504 -0.000035            0.000001    -0.000085
0.000715    -0.003918                  NaN      1.000000                    NaN
NaN                   NaN            NaN                   NaN      -0.000050
NaN       0.000090
smart_187_normalized        0.071719 -0.044238            0.397298    -0.521230
0.062075    -0.061517                  NaN           NaN              1.000000
-0.165096              0.035482      -0.015066              0.094061      -0.0543
75              0.094061      -0.054375
smart_187_raw              -0.002513  0.009072           -0.070568     0.062572
-0.001613     0.001540                  NaN           NaN             -0.165096
1.000000             -0.029712       0.004159             -0.015729       0.00766
2             -0.015729       0.007664
smart_188_normalized       -0.001367 -0.036524            0.079715    -0.025987
-0.001278     0.001295                  NaN           NaN              0.035482
-0.029712              1.000000      -0.003933              0.344726      -0.6377
34              0.344726      -0.637734
smart_188_raw              -0.008700  0.006620           -0.007481     0.005007
-0.017820     0.017709                  NaN           NaN             -0.015066
0.004159             -0.003933       1.000000             -0.009886       0.01241
5             -0.009886       0.012415
smart_197_normalized       -0.072106 -0.007484            0.857067    -0.003841
-0.008305     0.055798                  NaN           NaN              0.094061
-0.015729              0.344726      -0.009886              1.000000      -0.0954
46              0.640729      -0.095597
smart_197_raw              -0.001853  0.039558           -0.044346     0.012066
0.002043     0.001423                  NaN     -0.000050             -0.054375
0.007662             -0.637734       0.012415             -0.095446       1.00000
0             -0.105954       0.993962
```

```
smart_198_normalized       -0.039339 -0.014159           0.560889    -0.005163
0.017945     0.028147                  NaN           NaN                0.094061
-0.015729              0.344726      -0.009886              0.640729      -0.1059
54              1.000000      -0.108451
smart_198_raw              -0.001760  0.037546          -0.043418     0.012174
0.000525     0.001436                  NaN      0.000090               -0.054375
0.007664             -0.637734       0.012415             -0.095597       0.99396
2             -0.108451       1.000000
```




```
Strongest Correlation:
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Value</th> </thead> <tbody> <th>Feature_1</th> <th>Feature_2</th> <th>Correlation</th> <th>Absolute_Correlation</th> </tbody> <tbody></tbody>

smart_197_raw	
smart_198_raw	
0.993962	
0.993962	

```
Interpretation:
The strongest relationship is between smart_197_raw and smart_198_raw, with a cor
relation of 0.99.

Graph saved: output/visualizations\step_66_correlation_heatmap.png
```

In [12]:
```
# ============================================================
# STEP 67 — FAILURE STATUS COMPARISON
# ============================================================
```

```
print("\n" + "=" * 70)
print("STEP 67 — FAILURE STATUS COMPARISON")
print("=" * 70)

if target_column:

    feature = (
        selected_features[0]
    )

    grouped_means = (

        df

        .groupby(
            target_column
        )[feature]

        .mean()

        .to_frame(
            "Mean"
        )
    )


    show_and_save(

        grouped_means
        .reset_index(),

        "step_67_failure_status_means.csv",

        "STEP 67 — FAILURE STATUS MEANS"
    )


    plt.figure(
        figsize=(10, 6)
    )

    sns.boxplot(

        data=df,

        x=target_column,

        y=feature
    )

    plt.title(
        f"{feature} by Failure Status"
    )

    plt.xlabel(
        "Failure Status"
    )

    plt.ylabel(
        feature
```

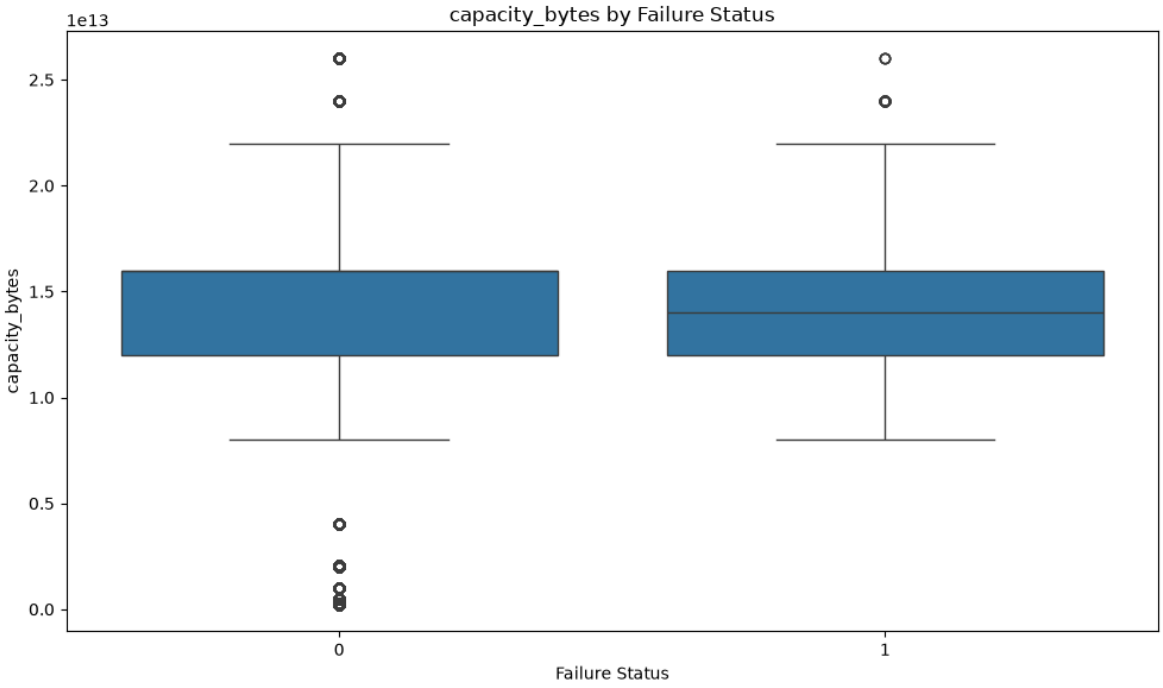
```
    )

    plt.tight_layout()

    graph_path = os.path.join(

        OUTPUT_FOLDER,

        "step_67_failure_status_comparison.png"
    )

    plt.savefig(

        graph_path,

        dpi=300,

        bbox_inches="tight"
    )

    plt.show()

    plt.close()


    highest_group = (
        grouped_means["Mean"]
        .idxmax()
    )


    interpretation = (

        f"The comparison shows how "
        f"{feature} varies between "
        f"failure-status groups. "
        f"The group '{highest_group}' "
        f"has the highest average value."
    )


    print(
        "\nInterpretation:"
    )

    print(
        interpretation
    )

    print(
        "\nGraph saved:",
        graph_path
    )


else:

    print(
        "Failure target was not detected."
    )
```

```
======================================================================
STEP 67 — FAILURE STATUS COMPARISON
======================================================================

======================================================================
STEP 67 — FAILURE STATUS MEANS
======================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>failure</th> <th>Mean</th> </thead> <tbody> <th>0</th> <th>1</th> </tbody> <tbody></tbody>

0	1.565091e+13
1	1.475768e+13

```
Saved CSV:
output/visualizations\step_67_failure_status_means.csv
```




```
Interpretation:
The comparison shows how capacity_bytes varies between failure-status groups. The
group '0' has the highest average value.

Graph saved: output/visualizations\step_67_failure_status_comparison.png
```

In [13]:
```python
# ============================================================
# STEP 68 — ALL GRAPH INTERPRETATIONS
# ============================================================

print("\n" + "=" * 70)
print("STEP 68 — ALL GRAPH INTERPRETATIONS")
print("=" * 70)

all_interpretations = []


if target_column:

    all_interpretations.append({
```

```
        "Step":
            60,

        "Graph":
            "Bar Chart",

        "Interpretation":
            f"The bar chart shows the distribution "
            f"of failure status. "
            f"'{target_counts.idxmax()}' has the "
            f"highest number of records."
    })


all_interpretations.append({

    "Step":
        61,

    "Graph":
        "Line Chart",

    "Interpretation":
        f"The line chart shows the variation "
        f"of {selected_features[0]} over "
        f"time or observations."
})


all_interpretations.append({

    "Step":
        62,

    "Graph":
        "Histogram",

    "Interpretation":
        f"The histogram shows the distribution "
        f"of {selected_features[0]}."
})


if len(selected_features) >= 2:

    all_interpretations.append({

        "Step":
            63,

        "Graph":
            "Scatter Plot",

        "Interpretation":
            f"The scatter plot shows the "
            f"relationship between "
            f"{selected_features[0]} and "
            f"{selected_features[1]}."
    })
```

```
all_interpretations.append({

    "Step":
        64,

    "Graph":
        "Box Plot",

    "Interpretation":
        f"The box plot shows the spread, "
        f"median, quartiles and possible "
        f"outliers of {selected_features[0]}."
})


if target_column:

    if len(
        df[target_column].unique()
    ) <= 6:

        all_interpretations.append({

            "Step":
                65,

            "Graph":
                "Pie Chart",

            "Interpretation":
                "The pie chart shows the "
                "proportion of each failure "
                "category."
        })


all_interpretations.append({

    "Step":
        66,

    "Graph":
        "Correlation Heatmap",

    "Interpretation":
        "The heatmap shows the correlation "
        "relationships among numerical features."
})


if target_column:

    all_interpretations.append({

        "Step":
            67,

        "Graph":
            "Failure Status Comparison",
```

```
        "Interpretation":
            f"The comparison shows how "
            f"{selected_features[0]} differs "
            f"between failure-status groups."
    })


interpretations_df = pd.DataFrame(
    all_interpretations
)

show_and_save(

    interpretations_df,

    "step_68_all_graph_interpretations.csv",

    "STEP 68 — ALL GRAPH INTERPRETATIONS"
)
```

```
======================================================================
STEP 68 — ALL GRAPH INTERPRETATIONS
======================================================================

======================================================================
STEP 68 — ALL GRAPH INTERPRETATIONS
======================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Step</th> <th>Graph</th> <th>Interpretation</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> </tbody> <tbody></tbody>

	Step	Graph	Interpretation
60		Bar Chart	The bar chart shows the distribution of failur...
61		Line Chart	The line chart shows the variation of capacity...
62		Histogram	The histogram shows the distribution of capaci...
63		Scatter Plot	The scatter plot shows the relationship betwee...
64		Box Plot	The box plot shows the spread, median, quartil...
65		Pie Chart	The pie chart shows the proportion of each fai...
66		Correlation Heatmap	The heatmap shows the correlation relationship...
67		Failure Status Comparison	The comparison shows how capacity_bytes differ...

```
Saved CSV:
output/visualizations\step_68_all_graph_interpretations.csv
```

Out[13]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Step</th> <th>Graph</th> <th>Interpretation</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> </tbody> <tbody></tbody>

60	Bar Chart	The bar chart shows the distribution of failur...
61	Line Chart	The line chart shows the variation of capacity...
62	Histogram	The histogram shows the distribution of capaci...
63	Scatter Plot	The scatter plot shows the relationship betwee...
64	Box Plot	The box plot shows the spread, median, quartil...
65	Pie Chart	The pie chart shows the proportion of each fai...
66	Correlation Heatmap	The heatmap shows the correlation relationship...
67	Failure Status Comparison	The comparison shows how capacity_bytes differ...

In [14]:
```
# ============================================================
# STEP 69 — VISUALIZATION SUMMARY
# ============================================================

visualization_summary = pd.DataFrame({

    "Step": [
        60,
        61,
        62,
        63,
        64,
        65,
        66,
        67
    ],

    "Visualization": [

        "Bar Chart",

        "Line Chart",

        "Histogram",

        "Scatter Plot",

        "Box Plot",

        "Pie Chart",

        "Correlation Heatmap",

        "Failure Status Comparison"
    ],

    "Purpose": [
```

```
        "Compare categories",
        "Identify trends",
        "Understand distributions",
        "Identify relationships",
        "Identify spread and outliers",
        "Show proportions",
        "Analyze correlations",
        "Compare features by failure status"
    ]
})

show_and_save(
    visualization_summary,
    "step_69_visualization_summary.csv",
    "STEP 69 — VISUALIZATION SUMMARY"
)
```

```
======================================================================
STEP 69 — VISUALIZATION SUMMARY
======================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Step</th> <th>Visualization</th> <th>Purpose</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> </tbody> <tbody></tbody>

60	Bar Chart	Compare categories
61	Line Chart	Identify trends
62	Histogram	Understand distributions
63	Scatter Plot	Identify relationships
64	Box Plot	Identify spread and outliers
65	Pie Chart	Show proportions
66	Correlation Heatmap	Analyze correlations
67	Failure Status Comparison	Compare features by failure status

```
Saved CSV:
output/visualizations\step_69_visualization_summary.csv
```

Out[14]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Step</th> <th>Visualization</th> <th>Purpose</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> </tbody> <tbody></tbody>

60	Bar Chart	Compare categories
61	Line Chart	Identify trends
62	Histogram	Understand distributions
63	Scatter Plot	Identify relationships
64	Box Plot	Identify spread and outliers
65	Pie Chart	Show proportions
66	Correlation Heatmap	Analyze correlations
67	Failure Status Comparison	Compare features by failure status

In [15]:
```
# ============================================================
# STEP 70 — FINAL VISUALIZATION SUMMARY
# ============================================================

print("\n" + "=" * 70)
print("STEP 70 — FINAL VISUALIZATION SUMMARY")
print("=" * 70)

png_files = [

    file

    for file in os.listdir(
        OUTPUT_FOLDER
    )

    if file.endswith(
        ".png"
    )
]


csv_files = [

    file

    for file in os.listdir(
        OUTPUT_FOLDER
    )

    if file.endswith(
        ".csv"
    )
]


final_summary = pd.DataFrame({
```

```
    "Output_Type": [

        "PNG Graphs",

        "CSV Results"
    ],

    "Number_of_Files": [

        len(png_files),

        len(csv_files)
    ]
})


show_and_save(

    final_summary,

    "step_70_final_visualization_summary.csv",

    "STEP 70 — FINAL VISUALIZATION SUMMARY"
)


print(
    "\nPNG Files:"
)

for file in sorted(
    png_files
):

    print(
        " -",
        file
    )


print(
    "\nCSV Files:"
)

for file in sorted(
    csv_files
):

    print(
        " -",
        file
    )


print(
    "\nOutput Folder:"
)

print(
    OUTPUT_FOLDER
```

```
)


print("\n" + "=" * 70)
print(
    "MEMBER 4 — STEPS 57–70 COMPLETED SUCCESSFULLY"
)
print("=" * 70)
```

```
======================================================================
STEP 70 — FINAL VISUALIZATION SUMMARY
======================================================================

======================================================================
STEP 70 — FINAL VISUALIZATION SUMMARY
======================================================================
```

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Output_Type</th> <th>Number_of_Files</th> </thead> <tbody> <th>0</th> <th>1</th> </tbody> <tbody></tbody>

PNG Graphs	8
CSV Results	14

```
Saved CSV:
output/visualizations\step_70_final_visualization_summary.csv

PNG Files:
 - step_60_bar_chart.png
 - step_61_line_chart.png
 - step_62_histogram.png
 - step_63_scatter_plot.png
 - step_64_box_plot.png
 - step_65_pie_chart.png
 - step_66_correlation_heatmap.png
 - step_67_failure_status_comparison.png

CSV Files:
 - step_59_selected_features.csv
 - step_60_bar_chart_data.csv
 - step_60_bar_chart_interpretation.csv
 - step_61_line_chart_result.csv
 - step_62_histogram_result.csv
 - step_63_scatter_result.csv
 - step_64_box_plot_result.csv
 - step_65_pie_chart_data.csv
 - step_65_pie_chart_interpretation.csv
 - step_66_correlation_matrix.csv
 - step_66_correlation_pairs.csv
 - step_67_failure_status_means.csv
 - step_68_all_graph_interpretations.csv
 - step_69_visualization_summary.csv

Output Folder:
output/visualizations

======================================================================
MEMBER 4 — STEPS 57–70 COMPLETED SUCCESSFULLY
======================================================================
```